Task 1 - Web App QA & Debug Report

Application RealWorld Svelte demo (Conduit) - https://realworld.svelte.dev/

Tested by Varsha

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Environment Public live demo, Chromium-based browser with DevTools. Manual functional testing.

Scope Sign-up, login, create / view / edit / delete article, logout.

1. Summary

Seven issues were found. Two are critical: the article page fails to load at all, and registering a second account with an existing username/email locks the original owner out. Several other symptoms (publish error, unreachable Edit/Delete controls) appear to be downstream of the article-page failure rather than separate bugs. Each issue lists what was directly observed and, separately, a suspected cause. Suspected causes are hypotheses: I had no access to server code or logs.

2. Bug Report

#	Title / Summary	Steps to Reproduce	Expected vs Actual	Severity	Suspected Cause
1	**Article page returns HTTP 500 (blocks read, edit, delete)** Evidence: Fig. 1, 3	1. Open https://realworld.svelte.dev/article/introduction-to-machine-learning-for-developers (or click any article from the feed). 2. Wait.	**Expected:** article content renders. **Actual:** bare "500 Internal Error" page after ~12 s. Only the page document is requested (500); no JS or API calls load. Edit and Delete buttons never appear because the page never renders.	**Critical**	Server-side render or upstream data fetch for the article hangs/errors and crashes the page build. Failure is before the app loads, so it is not a front-end bug. Needs server logs to confirm. Could also be environmental (shared demo backend).
2	**Duplicate registration succeeds and locks out the original account** Evidence: Fig. 2	1. Register account A (username + email). 2. Sign out and register again with the same username and email. 3. Sign in to account A.	**Expected:** second registration rejected with a field error. **Actual:** second account is created; original account can no longer be accessed - "Something went wrong."	**Critical**	No uniqueness constraint on username/email, so two records share identifiers; login lookup then fails and an unhandled error is shown as a generic message.
3	**Publishing shows HTTP 500 although the article is saved**	1. Log in, click New Post, fill in the form. 2. Click Publish. 3. Wait for the error, then refresh the feed.	**Expected:** success and redirect to the article. **Actual:** 500 error after loading; after refresh the article is in the feed.	**High**	Article is saved, then the redirect goes to the article page, which fails (issue 1). Likely the same root cause.
4	**Slow responses (~10-12 s)** Evidence: Fig. 3, 5	1. Open DevTools Network (no throttling). 2. Submit the sign-up form, or load an article URL.	**Expected:** response in about 1-2 s. **Actual:** register request took 10.60 s; failing article request took 11.96 s.	**Medium**	Slow backend or cold start on the shared demo; no timeout or loading feedback. Latency may also be what turns the article fetch into a 500 (issue 1).

#	Title / Summary	Steps to Reproduce	Expected vs Actual	Severity	Suspected Cause
5	**Generic error on failed login** Evidence: Fig. 2	1. Open Sign in. 2. Enter a valid email with a wrong password. 3. Submit.	**Expected:** clear message such as "email or password is invalid". **Actual:** full-page "Something went wrong."	**Medium**	Catch-all error handler does not map auth failures (401/422) to a user-facing message. The same message appears in issue 2.
6	**One-character password accepted** Evidence: Fig. 4, 5	1. Open Sign up. 2. Enter any username, a fake email and the password "a". 3. Submit.	**Expected:** password rules (minimum length) enforced. **Actual:** account created and user logged in.	**Medium**	No client-side or server-side password validation.
7	**No email verification**	1. Register with a made-up email address. 2. Check whether a verification step is requested.	**Expected (production):** email confirmed before use. **Actual:** account active immediately, no verification.	**Low**	Feature not implemented. Production-readiness gap rather than a defect; the RealWorld spec does not require it.

3. Root-Cause Analysis - Issue 2: Duplicate registration locks out the original account

When a second account is registered with the same username and email as an existing one, the application accepts it, and the original account can no longer be used: signing in shows "Something went wrong." This was reproduced by registering the same identifiers twice and then trying to sign in to the first account. The registration step is evidently not enforcing uniqueness on username or email, so two records now share the same identifiers. The most likely mechanism (a hypothesis, since I could not see server code or logs) is that the login lookup now matches two records where it expects one, which raises an unhandled error. A catch-all error handler then converts that into a generic message instead of a specific one. The same handler plausibly explains why a wrong password also shows "Something went wrong" (issue 5). To fix it, add UNIQUE constraints on email and username in the database, check for an existing user before inserting, and return a validation error (HTTP 422 in the RealWorld API spec) with a field-level message. Make login deterministic and return a clear "email or password is invalid" response, and merge or remove the existing duplicate rows. Finally, add a regression test that registers the same email twice and expects the second attempt to be rejected.

4. Test Scenario Results

Scenario	Result
Sign-up (new account)	Account created and auto-logged in (Fig. 5), but accepts a one-character password and took 10.6 s.
Sign-up (duplicate username/email)	Fail - accepted; original account locked out (issue 2).
Login with wrong password	Fail - generic "Something went wrong" (issue 5).
Create article	Partial - article saved, but 500 shown (issue 3).
View article (own or another user's)	Fail - HTTP 500 (issue 1).
Edit / delete article	Blocked - controls unreachable because the article page does not render.
Logout	Pass - returns the user to the home page.

5. Recommendations

1. Fix the article page server-side load first; it unblocks read, edit, delete and the post-publish redirect (issues 1, 3).
2. Enforce unique username and email in the database and in the sign-up handler; clean up existing duplicates (issue 2).
3. Map authentication failures to a clear message instead of a catch-all error page (issue 5).
4. Add password rules (minimum length at least) and, for production, email verification (issues 6, 7).
5. Add timeouts, loading states and monitoring for slow requests (issue 4).
6. Add regression tests: duplicate registration, wrong-password login, publish-then-open-article.

6. Limitations

No server logs or code were available, so all causes are hypotheses; observed behaviour is documented as observed. The target is a shared public demo, so the article 500 may be environmental rather than a code defect. A screenshot of the second (duplicate) registration itself was not captured; only the resulting login failure (Fig. 2). The DevTools console badge showed several errors during the sign-up flow (Fig. 4, 5); they were not analysed.

Appendix - Evidence Screenshots

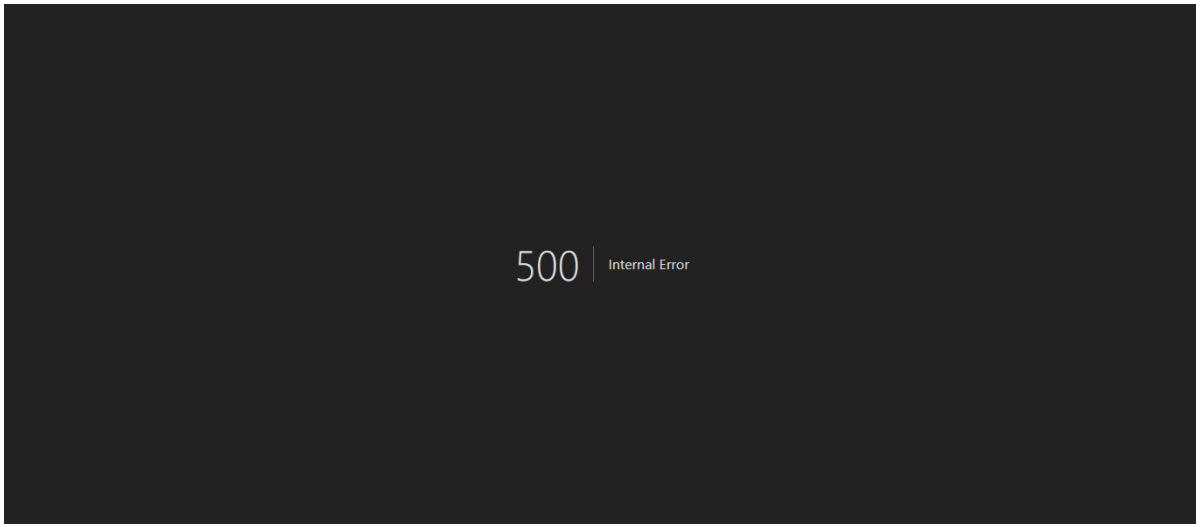


Fig. 1 - Opening an article URL returns a bare "500 Internal Error" page with no Conduit header or navigation.

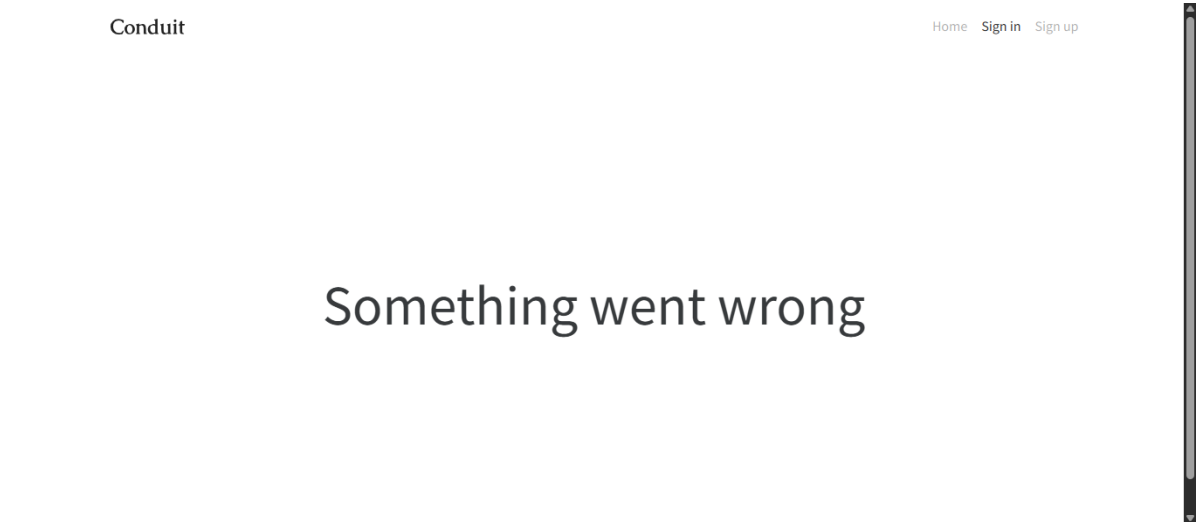


Fig. 2 - After registering a second account with the same username/email, signing in to the original account shows "Something went wrong."

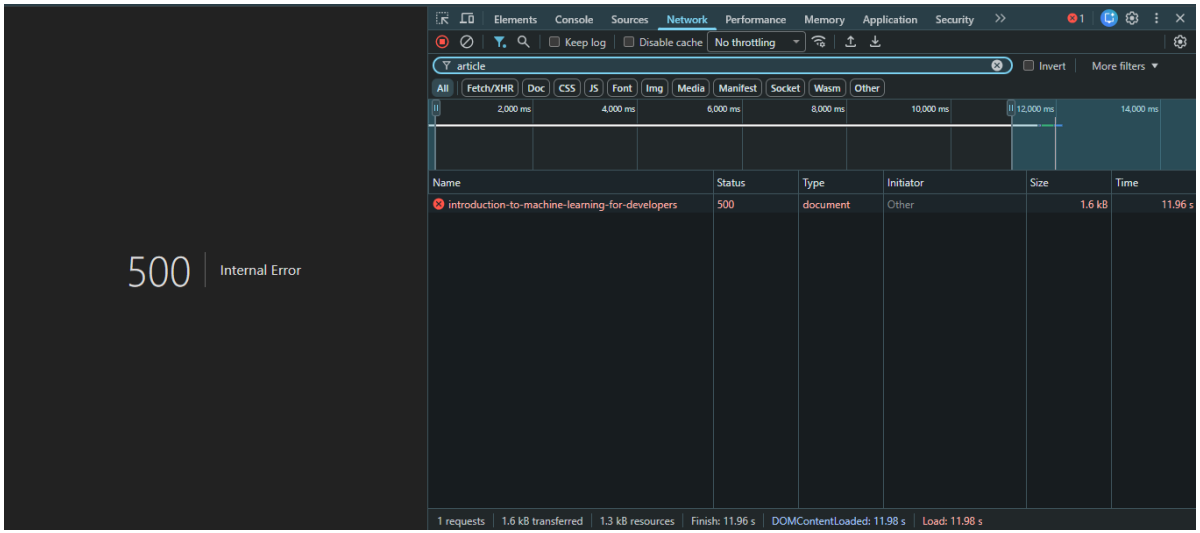


Fig. 3 - DevTools Network tab (Keep log on, no throttling): the only request is the page document for /article/introduction-to-machine-learning-for-developers, status 500, 11.96 s. No scripts or API calls load.

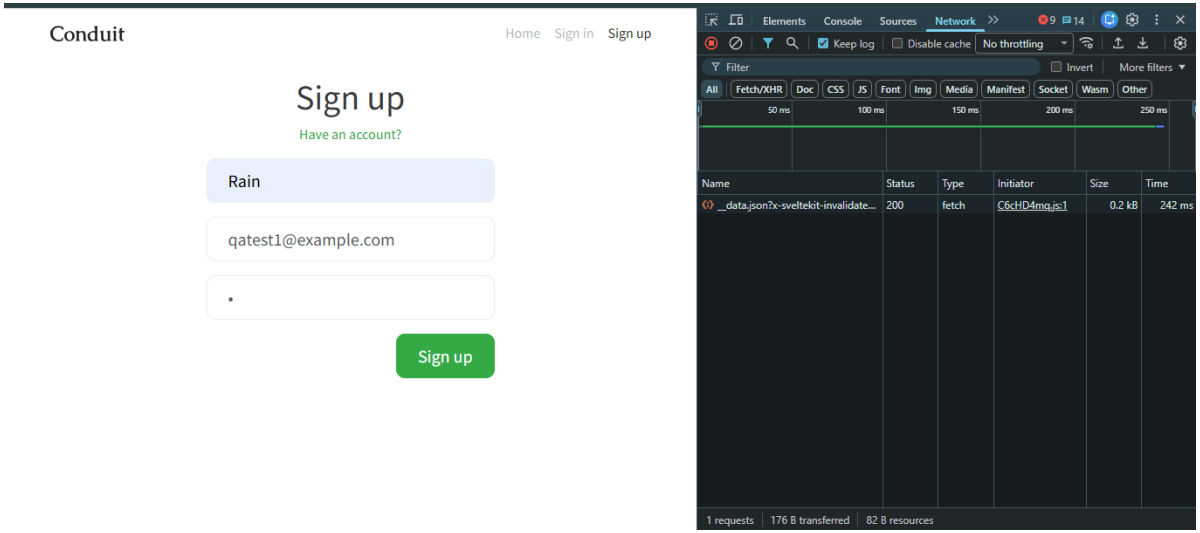


Fig. 4 - Sign-up form filled with a one-character password (throwaway account qatest1@example.com).

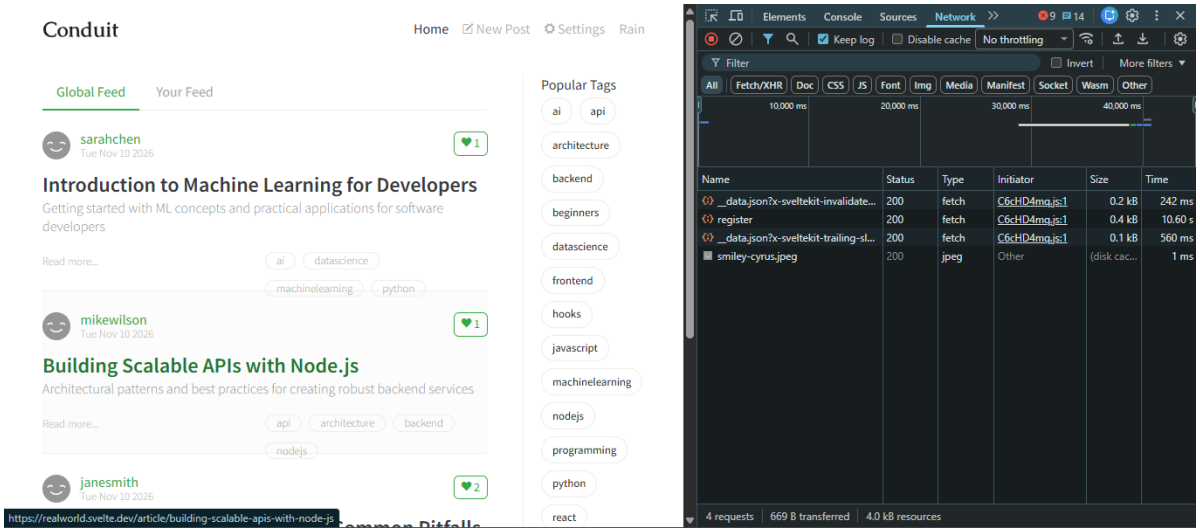


Fig. 5 - After submitting: account created and user logged in ("Rain" in header). The register request returned 200 and took 10.60 s.